

```

1a) if (grade >= 90) {
    System.out.println("Good job");
}
1b) if (number < 20 || number > 50) {
    System.err.println("Error: Number must be between 20 and 50.");
}
1c) if (y < 100) {
    y += 2;
}
2. if (num1 > num2) {
    System.out.println("The first number is larger.");
} else if (num2 > num1) {
    System.out.println("The second number is larger.");
} else {
    System.out.println("The numbers are equal.");
}

```

3a) Even first; after else, it's odd.

```

3b) switch (num % 2) {
    case 0:
        System.out.println("___ number"); // Even number case
        break;
    case 1:
    case -1:
        System.out.println("--- number"); // Odd number case
        break;
    default:
        break;
}

```

```

4a) int randomNumber = (int) (Math.random() * 50) + 1;
4b) int randomNumber = (int) (Math.random() * (100 - 20 + 1)) + 20;
4c) int randomNumber = (int) (Math.random() * (20 - 10 + 1)) + 10;

```

5. The third else if statement should only be else. It should not be else if. As it is the last statement,

6.a) true b) false c) false d) false e) true f) true g) false

8. a) false b) false c) false d) true e) false f) true g) true h) false i) true j) true

